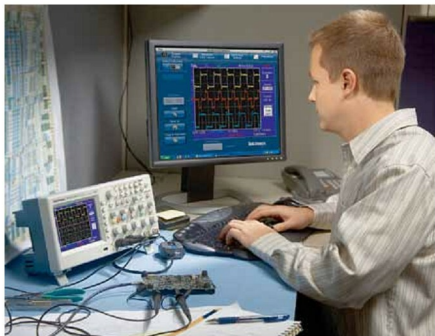


IS YOUR OSCILLOSCOPE SMART AND CONNECTED?

Higher analogue bandwidth, higher sampling rate, higher memory points and faster response are the biggest trends seen in oscilloscopes. But inbuilt support for connectivity to a computer or the network is the key specification that will drive the next-generation scopes



Tektronix TDS 2024B oscilloscope connected to network

■ SHWETA DHADIWAL BAID

Equipment vendors had foreseen that test and debug engineers would need to connect their equipment to the computer. Connectivity through various standards and modes has thus become the biggest trend in oscilloscopes today. As the computers continue to deliver higher performance and speed, they have become an attractive option for higher deliverables in test and measurement.

According to a Frost & Sullivan report, the oscilloscope market is primarily driven by constant technology innovations in the communication and computer industry. In 2009, the

world oscilloscope market generated revenues of \$987.7 million, which is expected to reach \$1513.4 million by 2014 as the demand for new products with enhanced features like PC connectivity along with Internet support increases.

The transition from analogue to digital in the communication and computer industry has given rise to the need for high-performance RF test tools, which has accelerated the demand for digital as well as PC-based oscilloscopes or digitisers. Users demand connectivity not only for waveform data transfer but also for control and remote access of the equipment. Thus Ethernet, universal serial bus (USB) and general-purpose interface

bus (GPIB) have become key specifications in next-generation oscilloscopes.

Modularity and virtual instrumentation is another technology trend seen in oscilloscopes, where the computer becomes the front-end screen while PXI or LXI ports enable the connectivity of oscilloscope hardware.

The phase of change

Oscilloscopes are the basic test and debug equipment found in every phase of electrical or electronic development. As consumer and industrial products become more intelligent with advanced embedded electronics, vendors aim at providing more and more features that are vital to supply products which are based on the new standards.

"Oscilloscopes have evolved from the traditional analogue scopes to high-performance digital oscilloscopes. Today, oscilloscopes have extremely high-speed analogue-to-digital converters (ADCs) and a huge memory of nearly 2 giga-points," informs Sanchit Bhatia, digital applications' consultant, Agilent Technologies. They have unprecedented analysis capabilities and more and more automated compliance software.

"Deeper memories, higher sampling rates, colour TFT and LCD wide-screen displays, and built-in options like function generator, DSO demonstrator and multichannel logic analyser are the major technological developments seen in oscilloscopes," shares Chandeeep Singh, CEO, Silicom Electronics.

In addition to the internal technology of the device, the development and enhancement of the probes play a